

NIKHIL H KULKARNI

SENIOR SOFTWARE ENGINEER

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SKILLS

- Java
- JavaScript
- Node.js
- React.js
- Spring Framework
- Spring Boot
- Jersey Framework
- Hibernate
- GraphQL
- REST
- SOAP
- JUnit
- PostgreSQL
- MySQL
- MongoDB
- DynamoDB
- Redis / RedisCache
- Relational Databases
- NoSQL
- Amazon Web Services (AWS)
- Microsoft Azure
- Apache Kafka
- MQTT
- Docker
- Jenkins
- Maven
- Git
- Agile
- Jira
- Wavefront

LIBRARIES

Redis npm lib for React-Native 4,816+ downloads

<https://www.npmjs.com/package/react-native-repubsub>

PATENTS

A Cargo Tracking System for Tracking a Cargo
WO2022037827

(PCT/EP2021/067474) · Filed Jun 8, 2021

Peer-to-peer sharing – self filed 2025
Utility patent pending

EDUCATION

Master's in Computer Science
GPA 4.0/4.0

Bachelor's in Computer Science
GPA 7.75 / 10

Diploma in Computer Science
GPA 7.5/10

WORK EXPERIENCE

Sr Software Engineer II (L6)

Intuit

Oct 2020 - Present

- **Architected** a **Redis** cache-based **microservice** to replace legacy late fee and reminder systems in production, resulting in a 2–3x improvement in processing time and enhancing overall **system performance**.
- Developed a SPAM detection engine using Drools (**rule-based system**), **leveraging telemetry** such as IP, age, and traffic flow to improve email filtering effectiveness by 70–80%.
- **Led** migration of monolithic **Kafka** events to a **Spring Boot microservices** architecture, analyzing legacy Java codebases and **RDBMS** schemas to accelerate modernization and enable future API migrations.
- **Implemented** a scalable **exponential back-off retry mechanism** for **AWS SQS** using Java, **OOP**, and **SOLID** principles, enabling configurable recovery strategies across distributed services.
- Refactored legacy print logic by developing **Thymeleaf**-based responsive **HTML** templates, enhancing PDF generation workflows and improving maintainability.
- Built a secure internal web tool (**Java + Spring Boot + React**) to query databases(**SQL reads**) without SSH tunneling, significantly reducing reconnection overhead; implemented **CORS** policies to support cross-origin requests from a decoupled frontend.
- Created a custom **Java annotation-based** version management system using **SOLID principles**, ensuring consistent version tracking across services and resolving a critical versioning issue in the legacy **monolith**.
- Enhanced observability by improving **Wavefront** and **Splunk dashboards**, supporting real-time telemetry analysis and faster incident response.
- Collaborated in **Agile teams** through code reviews, on-call rotations, and incident handling, driving secure, scalable, and resilient service delivery.
- Leveraged modern **DevOps** tools including **Kubernetes, Docker, Jenkins, Git, Maven, JUnit, and CI/CD pipelines** to implement TDD, **automate** testing, and streamline releases.
- **Authored** and maintained detailed **Confluence documentation**, managed multiple Maven repositories, and led **database** version upgrades to support platform consistency and compliance.

Full Stack Engineer

Mercedes-Benz R&D

Nov 2016–Sept 2020

- **Led** a team of 8–9 engineers in architecting a highly available, full-stack cloud infrastructure (**Vue.js, Node.js, Java-based microservices**) capable of supporting millions of connected devices with 99.9% availability.
- Developed a **React Native** package integrated with Redis to enable real-time notification exchange between **IoT** devices and **mobile apps**.
- Built a React Native mobile application to display live barcode scan data, enhancing visibility into IoT device interactions.
- Developed an telemetry system to track vans and other camera related data points using **kafka streams** and **elasticsearch**
- Developed an application to transfer data using the **LWM2M protocol** (similar to **MQTT**), supporting data ingestion to both **Azure and AWS cloud platforms**.
- Increased unit test coverage from 70% to 95% by writing robust **JUnit** and **Mockito** test cases, greatly improving system reliability and reducing production bugs.
- Installed and maintained **TensorFlow, CUDA**, and other **ML dependencies** on an **ARM processor** board, supporting edge-based machine learning workloads in an embedded environment.

Senior Software Engineer

Veg.Me

Feb 2016 - Jan 2017

- Built scalable backend services with **Node.js, GraphQL, and Redis** for real-time data handling.
- Automated image processing using **AWS Lambda, S3, and Python with CloudSearch** integration.
- Designed real-time pipelines using **AWS SQS, SNS, and Lambda** for content distribution.
- Migrated serverless infrastructure from **AWS to GCP, deploying via Cloud Functions** and App Engine.
- Developed reusable UI components with React Native, improving app modularity and speed of development.
- Used **Docker** to automate builds and streamline deployment across cloud environments.

Senior Software Engineer

Persistent Systems

Oct 2012 - Apr 2015

- Built billing and contract modules using **Spring, Struts, and Hibernate**.
- Developed **REST APIs** with **Spring MVC**; boosted test coverage to 95%.
- Led 2-member team as Scrum Master for contract and visitor modules.
- Created bulk ops APIs for user registration and onboarding.
- Enhanced **JMeter** with BeanShell for JSON parsing; **documented by IBM**.
- Built composite APIs to reduce load and improve response times.